MEMORIZE: none of this is given on the test

Properties of summation

(1) $\sum_{k=1}^{n} ca_k = c\sum_{k=1}^{n} a_k$ *constant out front*

(2) $\sum_{k=1}^{n}(a_k+b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$

(3) $\sum_{k=1}^{n}(a_k-b_k) = \sum_{k=1}^{n} a_k - \sum_{k=1}^{n} b_k$

(4) $\sum_{k=j+1}^{n} a_k = \sum_{k=1}^{n} a_k - \sum_{k=1}^{j} a_k$ ⋆⋆

use (4) when the lower index is not 1; j = start −1

Sum formulas (lower index must be 1)

(5) $\sum_{k=1}^{n} 1 = n$

(6) $\sum_{k=1}^{n} k = \dfrac{n(n+1)}{2}$

(7) $\sum_{k=1}^{n} k^2 = \dfrac{n(n+1)(2n+1)}{6}$

(8) $\sum_{k=1}^{n} k^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

Also know: $0! = 1,\ 1! = 1$ • Pascal's triangle (generate it, don't memorize it) • $\sin^2 t + \cos^2 t = 1$, $1+\tan^2 t = \sec^2 t$, $1+\cot^2 t = \csc^2 t$

Given on the test: the projectile equations $x = (v\cos\theta)t$, $y = (v\sin\theta)t - 16t^2 + h$. Just know how to use them.

The test (43 points, two days). Wed, Non-Calculator: 5 parametric curve questions + 1 sequence question. **Fri, Calculator:** 7 sequences & series questions + 1 parametric application (projectile). The calculator's Σ button is **blocked in test mode**: it's a homework check only. Show the formula work before you touch the calculator; that part is not optional.

P.8 Sequences; Summation Notation; the Binomial Theorem

1 Write the First Several Terms of a Sequence

A **sequence** is a function whose domain is the positive integers: plug in $n = 1,2,3,\ldots$ (never 0, never $\frac{1}{2}$). The rule for a_n is the **general term**.

EXAMPLE **Write the first six terms:** $\{S_n\} = \left\{(-1)^{n-1}\frac{2}{n}\right\}$

$a_1 = (-1)^0\frac{2}{1} = 2,\ a_2 = (-1)^1\frac{2}{2} = -1,\ a_3 = \frac{2}{3},\ a_4 = -\frac{1}{2},\ a_5 = \frac{2}{5},\ a_6 = -\frac{1}{3}$

Answer: $\boxed{2,\ -1,\ \frac{2}{3},\ -\frac{1}{2},\ \frac{2}{5},\ -\frac{1}{3}}$

Watch: *the answer is a list separated by commas. It asked for six, so no ellipsis. Reduce each term* $\left(-\frac{2}{4} = -\frac{1}{2}\right)$.

Pattern → *n*th term

1. Write 1, 2, 3, 4 above the terms.
2. Fractions: find the top and the bottom **separately**.
3. Alternating signs: use a power of −1. First term **positive**: $(-1)^{n+1}$. First term **negative**: $(-1)^n$.
4. Back-pocket rules: odd numbers $2n-1$, evens $2n$, squares n^2, powers b^{n-1} when the first term is 1.
5. Check $n = 1$ in your formula before you box it.

EXAMPLE $\overset{1}{2},\ \overset{2}{-4},\ \overset{3}{6},\ \overset{4}{-8},\ \overset{5}{10},\ldots$

numbers are $2n$; first term positive, so $\boxed{a_n = 2n(-1)^{n+1}}$ $(-1)^n$ *would make the first term* -2

EXAMPLE $\overset{1}{\frac{1}{3}},\ \overset{2}{-\frac{2}{9}},\ \overset{3}{\frac{3}{27}},\ \overset{4}{-\frac{4}{81}},\ldots$ **(test-level: fractions and alternating)**

top n, bottom 3^n, first term positive: $\boxed{a_n = (-1)^{n+1}\dfrac{n}{3^n}}$

Watch: *the test version is a little harder than the homework: expect different formulas on top and bottom plus alternating signs. $a_n = \ldots$ or braces { } both fine.*

2 Write the Terms of a Recursively Defined Sequence

A **recursive formula** gives the first term(s), then builds each term from the ones before it. The formula only starts where it says ($n \geq 2$).

EXAMPLE $s_1 = 1,\ \ s_n = 4s_{n-1}$. **Write the first five terms.**

$s_2 = 4s_{2-1} = 4s_1 = 4(1) = 4,\ \ s_3 = 4s_2 = 16,\ \ s_4 = 64,\ \ s_5 = 256$ Answer: $\boxed{1,\ 4,\ 16,\ 64,\ 256}$

EXAMPLE $u_1 = 1,\ u_2 = 1,\ u_n = u_{n-2} + u_{n-1}$

$u_3 = u_1 + u_2 = 2,\ \ u_4 = u_2 + u_3 = 3,\ \ u_5 = 5$ $\boxed{1,\ 1,\ 2,\ 3,\ 5}$ *the Fibonacci sequence*

Watch: $4s_{n-1}$ *means 4* times *the previous term. The subscript tells you which term, it is not a number to multiply.*

3 Use Summation Notation

$\sum_{k=1}^{n} a_k$ means *add* the terms a_k from $k = 1$ (the **index of summation** starts at the lower index) to $k = n$.

EXAMPLE **Write out the sum.**

$$\sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

$$\sum_{k=1}^{n} k! = 1! + 2! + 3! + \cdots + n!$$

first few terms, ellipsis, last term. Don't solve.

EXAMPLE **Express using summation notation.**

$$1^2 + 2^2 + \cdots + 9^2 = \sum_{k=1}^{9} k^2$$

$$2 + \tfrac{3}{2} + \tfrac{4}{3} + \cdots + \tfrac{35+1}{35} = \sum_{k=1}^{35} \frac{k+1}{k}$$

lower index, upper index, general term

Watch: *the letter on the bottom and inside must match (k and k, not k and n). In the second example, k starts at 1, not 2: $k = 1$ has to give the first term, 2.*

4 Find the Sum of the First n Terms of a Sequence

1. **Lower index not 1? Red flag.** Fix it first with (4). Start at $k = 6$ means $j = 5$. "Don't pass go."
2. Split sums and differences: (2), (3). Pull constants out: (1).
3. Write each formula **from memory** with n substituted. **This is how far you must get before the calculator.**
4. Type that line into the calculator exactly as written. Cite the numbers (1)–(8) next to each step.

EXAMPLE $\sum_{k=1}^{100} \frac{1}{2}$

$\frac{1}{2}\sum_{k=1}^{100} 1$ *(1)*

$\frac{1}{2}(100) = \boxed{50}$ *(5)*

EXAMPLE $\sum_{k=1}^{5} 3k$

$3\sum_{k=1}^{5} k$ *(1)*

$3\left(\frac{5(6)}{2}\right) = \boxed{45}$ *(6)*

EXAMPLE $\sum_{k=1}^{10} (k^3 + 1)$

$\sum_{k=1}^{10} k^3 + \sum_{k=1}^{10} 1$ *(2)*

$\left[\frac{10(11)}{2}\right]^2 + 10$ *(8), (5)*

$\boxed{3035}$

EXAMPLE $\sum_{k=6}^{20} 4k^2$

$4\sum_{k=6}^{20} k^2$ *(1)*

$4\left[\sum_{k=1}^{20} k^2 - \sum_{k=1}^{5} k^2\right]$ *(4): lower index 6, so $j = 5$*

$4\left[\frac{20(21)(41)}{6} - \frac{5(6)(11)}{6}\right]$ *(7). Stopping point, now the calculator*

$4[2870 - 55] = \boxed{11260}$

Watch: (8) *squares the* whole *bracket.* $\sum 1 = n$ *(adds n ones) is not* $\sum k$.

5 Use the Binomial Theorem

The binomial coefficient formula is **not given** and you don't need it. Use **Pascal's triangle**; the directions say Binomial Theorem, so multiplying it out loses credit.

1. Write the blanks: first term's powers count **down** from n to 0, second term's count **up** from 0 to n.
2. Fill the blanks with row n.
3. Simplify each term.

$n = 0$: 1
$n = 1$: 1 1
$n = 2$: 1 2 1
$n = 3$: 1 3 3 1
$n = 4$: 1 4 6 4 1
$n = 5$: 1 5 10 10 5 1
$n = 6$: 1 6 15 20 15 6 1
$n = 7$: 1 7 21 35 35 21 7 1
$n = 8$: 1 8 28 56 70 56 28 8 1

outsides are 1; each inside number is the sum of the two above it. Expect n up to 7 or 8.

EXAMPLE $(x+2)^5$

$1x^5(2)^0 + 5x^4(2)^1 + 10x^3(2)^2 + 10x^2(2)^3 + 5x(2)^4 + 1x^0(2)^5$

$\boxed{x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32}$

EXAMPLE $(2x-3)^4$ **row 1 4 6 4 1, first term** $2x$**, second term** -3

$1(2x)^4(-3)^0 + 4(2x)^3(-3)^1 + 6(2x)^2(-3)^2 + 4(2x)^1(-3)^3 + 1(-3)^4$

$\boxed{16x^4 - 96x^3 + 216x^2 - 216x + 81}$

Watch: *signs alternate only when the second term is negative: the minus sign goes with it, and an* ***odd*** *power of a negative is negative, so signs alternate.* $(2x)^4 = 16x^4$*, not* $2x^4$*. Calculator is fine for the arithmetic when* $a > 2$.

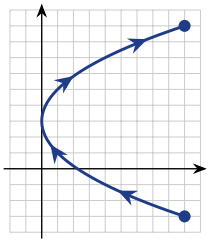
Parametric Equations, Graphs, and Applications

Graphing a Plane Curve Defined Parametrically

1. **Make a t-chart first**, always. Use only as many t values as it takes to see the shape.
2. **Brackets** $[a,b]$: plug in both endpoints and draw **solid dots** there. **Parentheses** (a,b): the endpoints are not used; pick t values inside.
3. **Direction**: start at the first t and follow the t-chart in order. Small arrows along the curve.
4. **Direction arrows only when the question asks. End arrows** only when the curve keeps going (t in $(-\infty,\infty)$, or it runs off the grid). Closed interval: dots, no end arrows.
5. Asymptotes are optional; if you draw them, dash them. No domain written means all real x.

EXAMPLE $x = t^2,\ y = 2t+3$, **for** t **in** $[-3,3]$

t	x	y
-3	9	-3
-2	4	-1
-1	1	1
0	0	3
1	1	5
2	4	7
3	9	9



starts at $(9,-3)$ when $t=-3$, ends at $(9,9)$. Solid dots, no end arrows.

Finding an Equivalent Rectangular Equation

1. **Pick the equation where solving for t gives one answer** (no $\pm$). Substitute into the other.
2. **Trig curves**: solve each for the trig function, then use the identity.
   $\sin,\cos \to \sin^2 t+\cos^2 t = 1 \quad \tan,\sec \to 1+\tan^2 t = \sec^2 t \quad \cot,\csc \to 1+\cot^2 t = \csc^2 t$
3. **State the domain**: "for x in $[a,b]$" using the x-values from the t-chart (watch turning points, like $x=t^2$).
4. Any form is fine once t is gone; no need to multiply it out. **SIMPLIFIED** in the question means get y alone. Restrictions are always on x (the domain).

EXAMPLE $x = t+2,\ y = t^2$, t **in** $[-1,1]$

$t = x-2$ *one answer*

$y = (x-2)^2$ *substitute*

$-1 \le t \le 1 \Rightarrow 1 \le x \le 3$ *domain from t*

$\boxed{y = (x-2)^2 \text{ for } x \text{ in } [1,3]}$

EXAMPLE $x = 2\sin t,\ y = 3\cos t$, t **in** $[0,2\pi]$

$\sin t = \frac{x}{2},\ \cos t = \frac{y}{3}$

$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ $\sin^2 t+\cos^2 t = 1$

$\boxed{\frac{x^2}{4}+\frac{y^2}{9}=1}$ *full ellipse: no restriction*

How far to simplify (sec/tan or csc/cot hyperbolas)

- t **restricted** so only one branch is traced: solve for y and keep the sign of that branch. Never write "$y \geq \ldots$".

  $x = 3\tan t,\ y = 2\sec t,\ t$ in $(-\frac{\pi}{2}, \frac{\pi}{2})$: $\frac{y^2}{4} - \frac{x^2}{9} = 1 \Rightarrow \boxed{y = 2\sqrt{\frac{x^2}{9} + 1}}$ *sec > 0 here, so top branch only;* $2\sqrt{1 + \frac{x^2}{9}}$ *or* $\sqrt{4 + \frac{4x^2}{9}}$ *also fine*
- t **not restricted**, both branches (the whole graph): done at "$= 1$".

  $x = 2\sec t,\ y = \tan t$: $\boxed{\frac{x^2}{4} - y^2 = 1}$ *also right:* $y = \pm\sqrt{\frac{x^2}{4} - 1}$*; never* $+\sqrt{}$ *alone*
- **One branch of a left-right hyperbola**: the domain can cut it, so "$\frac{x^2}{4} - y^2 = 1$ for x in $[2, \infty)$". Never "for y in".
- **The rule:** whole graph, nothing else to do. Part of it and the domain can cut it, say the domain. The domain can't (top or bottom of an up-down hyperbola), solve for y and take a sign.
- **Small piece of a branch:** solve for y *and* add "for x in". Never "for y in".
- **SIMPLIFIED** in the question: all the way, y alone ($y = -\frac{1}{3}x^2$ shows it goes through $(0,0)$). No "simplify": any form with t gone; multiplying out further just wastes time.
- **Make a t-chart** for every hyperbola: it shows which branch and which way it runs ($x = 4\cot t,\ y = 3\csc t$ on $(0, \pi)$ runs right to left).

Watch: *reciprocal curves: $x = t + 2,\ y = \frac{1}{t+2},\ t \neq -2$ gives $y = \frac{1}{x}$; "for all $x \neq 0$" is good practice, though it comes with the equation. Square roots: $x = \sqrt{t}$ means $x \geq 0$, so state "for x in $[0, \ldots]$".*

Finding Alternative Parametric Equation Forms

EXAMPLE $y = (x-2)^2 + 1$ ***Don't forget to include the parameter!***

Option I: $x = t,\ y = (t-2)^2 + 1,\ t$ in $(-\infty, \infty)$
Option II: $x = t + 2,\ y = t^2 + 1,\ t$ in $(-\infty, \infty)$

Give both $x =$ and $y =$; $t = 4 - x$ is not parametric form. Option I: replace x with t. Option II: pick $x = t + h$ so the inside becomes just t. Not in vertex form? Pick any shift $x = t + h$, substitute, simplify. Stop once t is in; expanding is a waste of time. Restricted t? Write that interval instead.

Applications of Parametric Equations (Friday, calculator)

1. Write the equations with the numbers: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$ (from the ground: $+0$). The **degree symbol** is required.
2. Calculator: **degree** mode, Graph $\rightarrow$ Parametric, type $x_1(t)$ and $y_1(t)$ *exactly*. Write the setup on the test.
3. Graph doesn't come back down? Raise $t$max (it defaults to 2π).
4. **Max height**: trace to the top, read y (Analyze, Maximum doesn't work on parametric graphs). **Time in the air**: trace to where $y \approx 0$, read t. **Distance**: read x at that same point.
5. Precision: trace close, change trace step to 0.01, then 0.001. Get y as close to 0 as you can. Thousandths.
6. Answer with units: "$y \approx 26.258$ ft," "$t \approx 2.512$ s," "$x \approx 126.707$ ft" is enough.

Don't burn 20 minutes on the calculator part.

Where points get lost

- -1: arithmetic slip while simplifying $x = -2t - 3,\ y = -\frac{4t^2}{3} - 4t - 3$. Correct: $y = -\frac{1}{3}x^2$ for x in $[-3, 3]$. Check one t-chart point in your final equation.
- -1: $x = 2\sec t,\ y = \tan t$ (no interval) answered $y = +\sqrt{\ldots}$. Leave it as $\frac{x^2}{4} - y^2 = 1$ (whole graph) or keep $\pm$.
- No points, but time: expanding $y = -2(t-4)^2 + 3$ isn't needed ("can stop here"). Units: speed in ft/s means answers in **feet**.

Key: t-chart = table of t, x, y values. (1)–(8) = the summation properties and formulas in the Memorize box.